Supplementary Material for "News Publisher Reliability from Article Collections: A Publisher-Disjoint Evaluation"

This supplementary material accompanies the main manuscript submitted to ECIR 2027, "IR for Good" track.

It reports in full the error and bias audit behind the main paper's conditions for responsible use, together with the experimental detail needed to reproduce them.

1 Model performance

Figure 1 shows the confusion matrices for each fold and for the average.

2 Publisher-level Results

Figure 2 reports the publisher-level confusion matrices obtained after aggregating article-level predictions by majority voting.

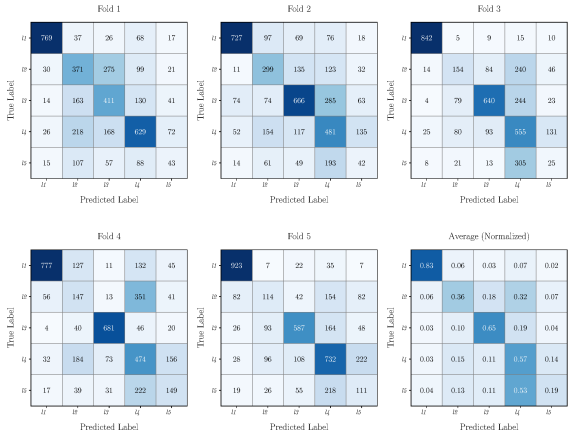
Most errors are local. The most frequent confusion is l_5 (High Credibility) predicted as l_4 (Generally Credible) for BERT, Llama 3 and Mistral, and the second most frequent for RoBERTa. As observed at the article level, distinguishing l_5 from l_4 is inherently difficult because l_5 represents a strict ceiling (score = 100) while l_4 covers the immediately lower high-credibility range (75–99). Misclassifying these two adjacent levels constitutes a minor severity error, as both indicate high reliability. Errors of two levels are nonetheless common, 39 to 46 publishers per model, and mostly fall between l_2 and l_4 in either direction; errors of three or more levels affect 10 to 17 publishers.

One caveat applies to every publisher-level number reported here and in the main paper. Aggregation is a majority vote, and ties are resolved deterministically in favour of the *lowest* reliability level. This affects a small but non-negligible share of publishers – 13 of 439 for BERT, 20 for RoBERTa, 15 for Llama 3 and 11 for Mistral – and always in the same direction, so it is a mild systematic pull toward lower predicted reliability rather than a neutral rule.

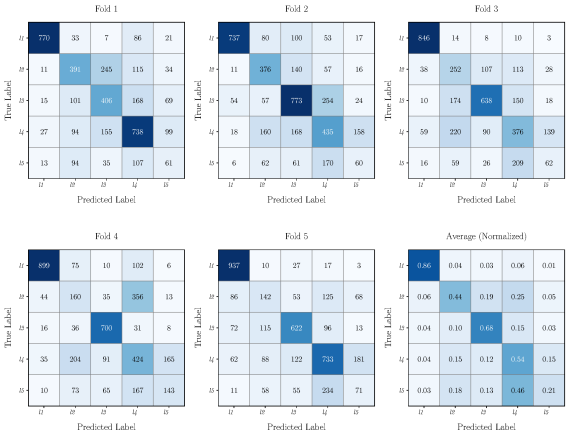
3 Political leaning

Figure 3 reveals why the Chi-square test rejects the hypothesis of independence: the distribution of errors varies between moderate and extreme political groups, with notable differences across models. The matrices for 'Slightly Right' and 'Non-labelled' orientations exhibit strong diagonal dominance across all architectures. For the former (the largest group, $N = 7490$), Mistral, RoBERTa, and BERT achieve comparable accuracies of 69.0%, 68.4%, and 67.6%, respectively, with Llama 3 at 66.4%. The domain counts (annotated as 'Dom' in the cells) show that this performance is robust across a large number of unique sources.

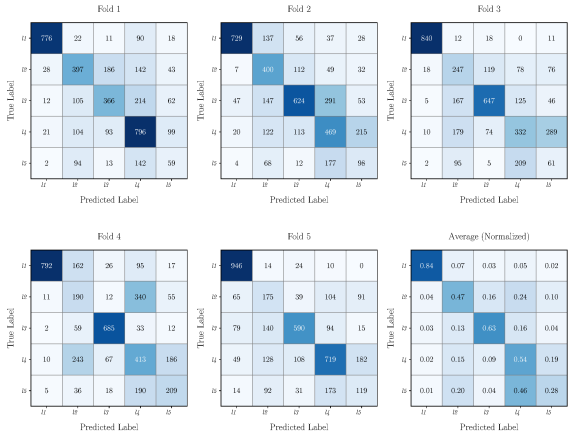
In contrast, the 'Far Left' ($N = 1015$) and 'Far Right' ($N = 2142$) matrices display high off-diagonal scattering and significantly lower global accuracy. For 'Far Left', all models perform



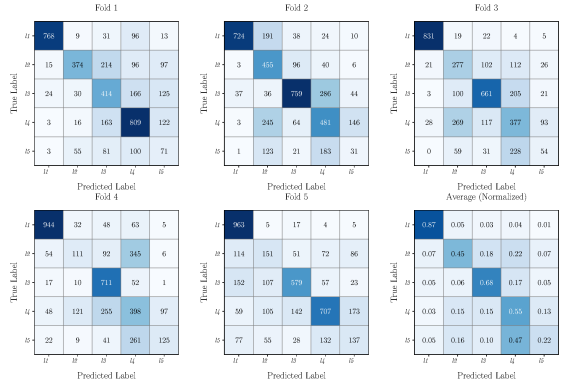
(a) BERT



(b) RoBERTa

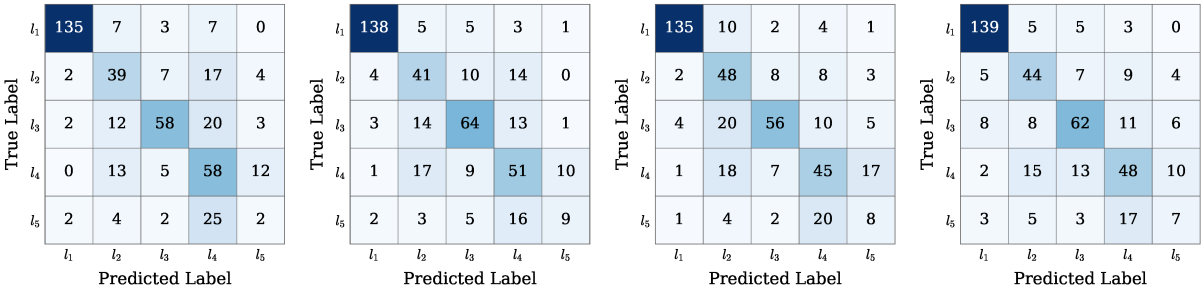


(c) Llama 3



(d) Mistral

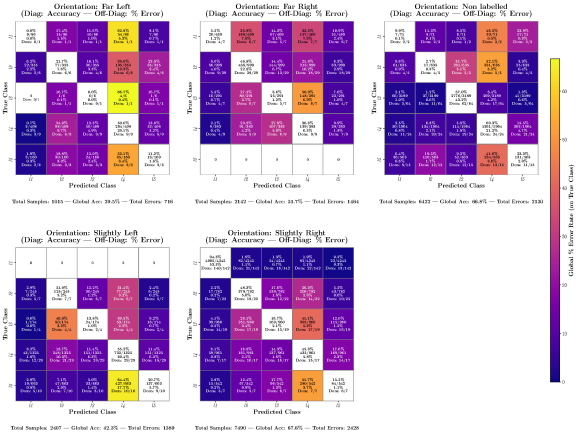
Figure 1: Confusion matrices showing the models' performance across five cross-validation folds and their overall average. The rows correspond to the true labels and the columns to the predicted ones.



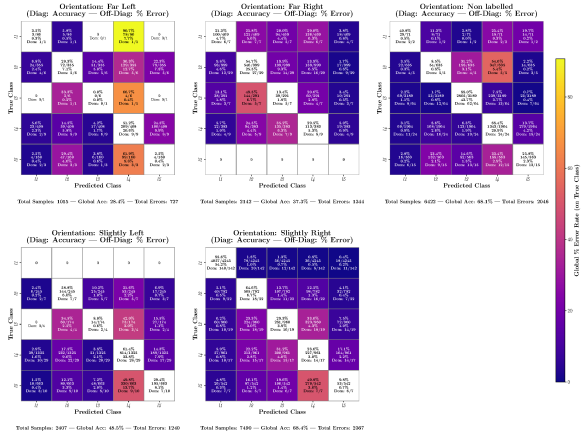
(a) BERT (b) RoBERTa (c) Llama 3 (d) Mistral

Figure 2: Publisher-level confusion matrices for BERT, RoBERTa, Llama 3, and Mistral. Rows indicate true reliability levels, columns indicate predicted levels.

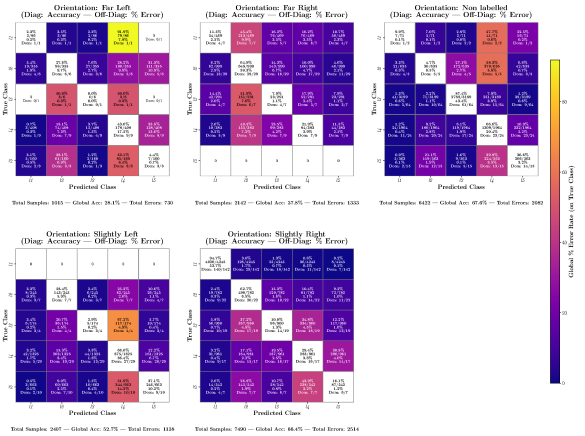
poorly with accuracies between 28.1% and 29.5% for the three smaller models and 36.5% for Mistral. Visual inspection confirms a systematic directional bias: predictions frequently land in the upper triangle of the matrix, confirming that models overestimate the reliability of these sources.



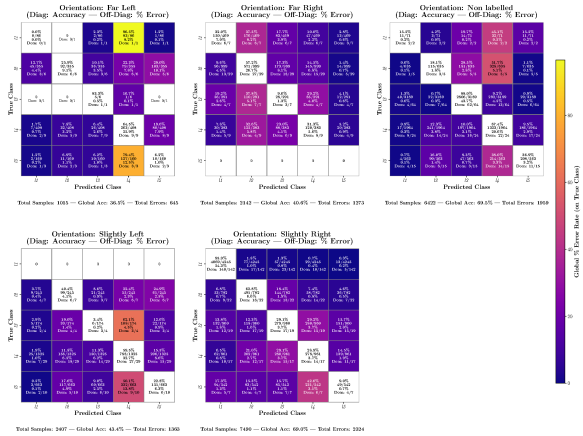
(a) BERT



(b) RoBERTa



(c) Llama 3



(d) Mistral

Figure 3: Confusion matrices for political orientation across the four models. The rows correspond to the true labels and the columns to the predicted ones. The cells show class-wise percentages and counts, dataset-level percentages and contributing domains, and the colour represents the error rate per true class.

The 'Far Right' category shows one of the largest differences between the models, second only to 'Slightly Left'. Mistral, Llama 3, and RoBERTa achieved 40.6%, 37.8%, and 37.3%, respectively, while BERT dropped to 31.7%.

Domain metrics further clarify that errors in these extreme categories are driven by a small set of outlets. Unlike the diverse 'Slightly Right' group, the 'Far Left' and 'Far Right' categories contain fewer unique domains e.g., *Dom: 1/1* or *Dom: 5/7*, yet these specific outlets consistently trigger model errors. The 'Slightly Left' matrix confirms the specific discrediting pattern discussed in the main manuscript, where models tend to underestimate the reliability of slightly left-leaning sources. A substantial cluster of misclassifications appears in the lower triangle, below the diagonal. This visually demonstrates that models penalize these articles, assigning them lower reliability labels compared to the ground truth, a behavior consistent across all four architectures.

We stress that these matrices are descriptive. Because political leaning and reliability are both properties of the publisher, and are themselves associated in the media ecosystem, the per-leaning error rates visible here cannot on their own separate a political effect from a reliability effect. The main paper addresses this with publisher-clustered tests that control for the true reliability level.

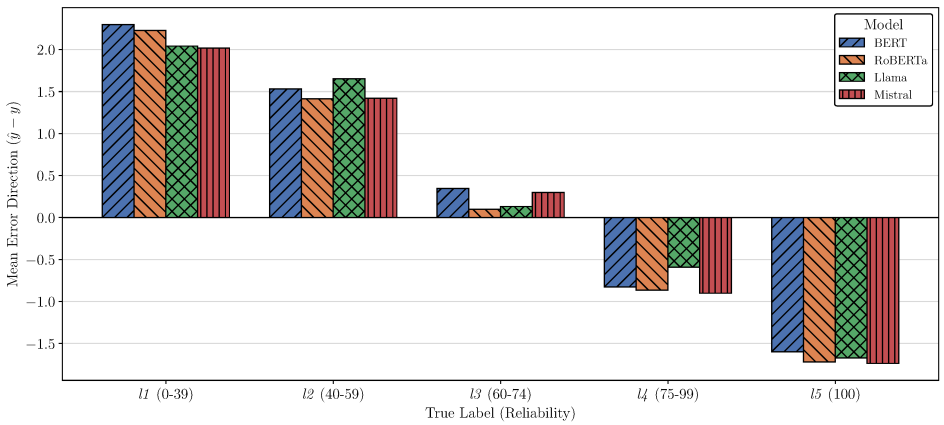


Figure 4: Mean Error Direction by true reliability level. Positive values indicate overestimation of reliability, negative values underestimation.

4 Error Direction at the Article Level

The main paper reports the per-class error rate; here we report the companion analysis of the *direction* of those errors, measured by the Mean Error Direction (MED), the average signed difference between predicted and true reliability labels over misclassified instances. Positive values indicate overestimation of reliability, negative values underestimation.

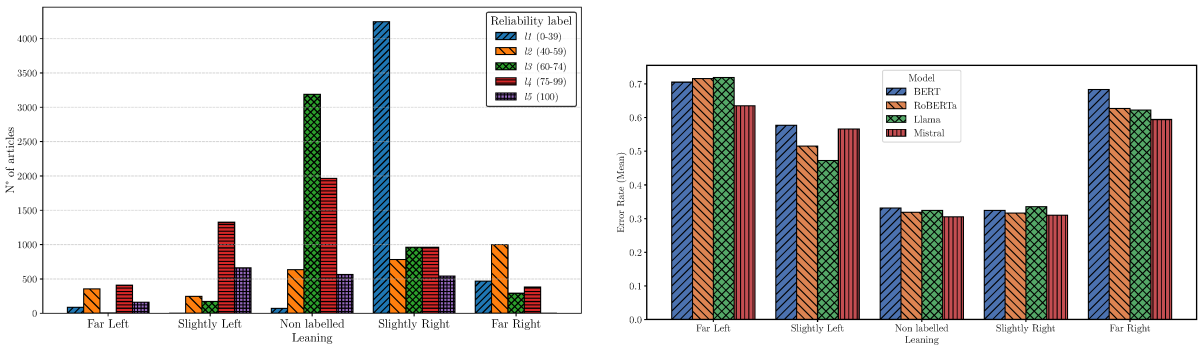
Figure 4 plots the Mean Error Direction (MED) to quantify the prediction bias. MED is the average signed difference between predicted and true reliability labels over the set of misclassified instances, correctly classified ones being excluded from the average. We observe a tendency to regress towards the center. We attribute this to two structural factors:

- **Boundary Constraints** (l_1 and l_5). Errors at the extremes are naturally constrained: models can only overestimate l_1 and underestimate l_5. However, the MED in l_5 is strongly negative, around -1.7 (BERT -1.60, RoBERTa -1.72, Llama -1.67, Mistral -1.74), and the one in l_1 strongly positive, around $+2.1$. The confusion matrices reveal that the vast majority of misclassifications for l_5 are concentrated in adjacent l_4 (Section 1). This confirms that the high error rate is primarily due to the models struggling to distinguish between perfect (100) and generally credible (75–99) reliability, likely because there is little textual distinction between these high-standard categories. The magnitude beyond -1.0 indicates a minor tendency to misclassify some l_5 instances into lower tiers as well, but the bulk of the errors consists of labeling l_5 as l_4.

- **The 'Safe Harbor' Effect (l_3).** Class l_3 shows the smallest Mean Error Direction of all levels, between 0.10 and 0.35 across models. l_3 acts as a gravitational center for uncertainty: it absorbs a significant portion of misclassifications from the lower-reliability l_2 and from the higher-reliability l_4. Since the model overestimates l_2 (pushing it up to 3) and underestimates l_4 (pushing it down to 3), these opposing error vectors cancel each other out, resulting in a net directional bias of zero.

5 Political Leaning: Distribution and Error Rates

The main paper reports the directional bias by political leaning and the associated significance tests, and it makes the disclosure of per-leaning error rates a condition for responsible deployment. Figure 5a gives the composition of the corpus, and Figure 5b the per-leaning error rates behind those tests. Readers can therefore inspect the disparity rather than take it on trust.



(a) Reliability labels across political leanings.

(b) Error rate by leaning and model.

Figure 5: Composition of the corpus by political leaning (left) and the resulting per-leaning error rates (right).

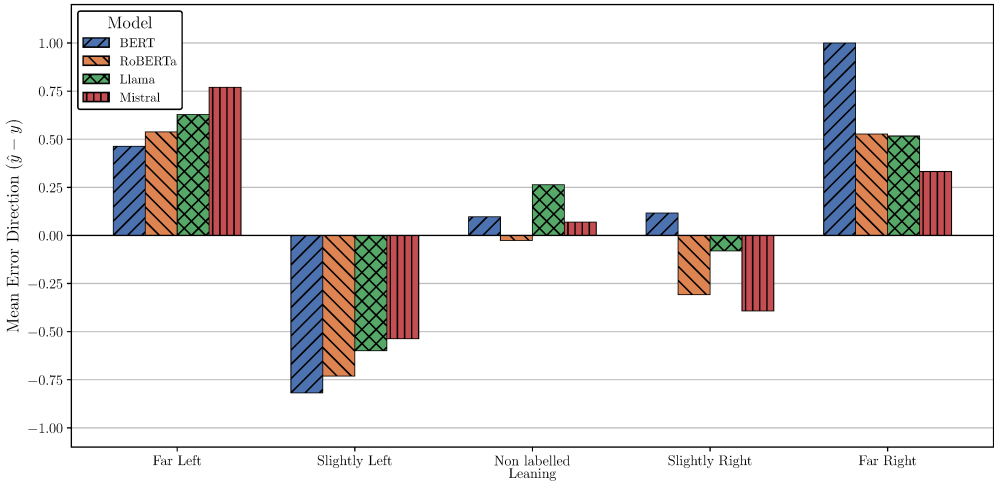


Figure 6: Mean Error Direction by political leaning. Positive values indicate overestimation of reliability (prediction above the reference level), negative values underestimation. The values are discussed in the main paper.

The distribution is markedly imbalanced: reliability levels are not uniformly represented within each political category, which is a structural property of the media ecosystem rather than of our sampling. This is why error rates are computed class-conditionally in the main paper, making them independent of category frequency. Mistral deviates from the other three models, showing the fewest errors in both extreme categories, while its error rate on Slightly Left is close to BERT's and above those of RoBERTa and Llama, and it performs comparably on Non labelled and Slightly Right.

6 Robustness of the Bias and Performance Analyses

The main paper reports tests that treat each article as an independent observation. Political leaning and reliability are publisher-level attributes, and every article of an outlet carries the same value. This section therefore repeats the analyses at the unit on which the labels actually live.

The contingency tables behind the test. The χ^2 test of independence reported in the main paper is computed on the table that crosses political orientation with classification outcome:

five orientations against two outcomes, over all 19,476 articles. Table 1 gives it for each model. With $c = 2$ the factor $\min(r-1, c-1)$ equals one, so Cramér's V is simply $\sqrt{\chi^2/N}$.

	BERT		**RoBERTa**		**Llama 3**		**Mistral**	
Leaning	correct	wrong	correct	wrong	correct	wrong	correct	wrong
Far Left	299	716	288	727	285	730	370	645
Slightly Left	1018	1389	1167	1240	1269	1138	1044	1363
Non labelled	4292	2130	4376	2046	4340	2082	4463	1959
Slightly Right	5062	2428	5123	2367	4976	2514	5166	2324
Far Right	678	1464	798	1344	809	1333	869	1273
Total	11349	8127	11752	7724	11679	7797	11912	7564

Table 1: Political orientation against classification outcome, article level, one pair of columns per model. These are the tables the χ^2 statistics and the effect sizes of the main paper are computed from.

Independence at the publisher unit. Table 2 repeats the χ^2 test of independence with one observation per publisher: was the aggregated prediction for that outlet correct? The 439 publishers replace the 77,904 articles, and no expected count falls below five. The association between political leaning and correctness survives for every model, and the effect size is larger than at the article level.

Model	n	χ^2	p**-value**	**Cramér's V**
BERT	439	44.84	4.3×10^{-9}	0.320
RoBERTa	439	18.83	8.5×10^{-4}	0.207
Llama 3	439	23.88	8.4×10^{-5}	0.233
Mistral	439	24.06	7.8×10^{-5}	0.234

Table 2: Political leaning vs. classification correctness at the publisher unit, 4 degrees of freedom. Compare with the article-level table of the main paper.

Leaning or reliability? Reliability levels are unevenly distributed across political leanings, and some levels are harder than others. An apparent political effect could therefore be a reliability effect in political guise. For each model we fit a logistic regression of the article-level error indicator on political leaning, we enter the true reliability level as a control, and we cluster the standard errors on the publisher. Table 3 reports the result against the non-labelled reference group. Far-left and far-right outlets keep four to six and a half times the odds of error of non-labelled ones, with the reliability level already accounted for. The confidence intervals are wide, and deliberately shown: with several cells holding fewer than ten publishers, the direction of the effect is better established than its size.

Uncertainty on the publisher-level scores. Table 4 reports 95% percentile intervals from a *cluster* bootstrap, which resamples whole publishers rather than individual articles. The intervals of the four models overlap on both metrics. At this sample size the differences between them, 0.69 against 0.68 on accuracy, are not distinguishable from sampling variation.

Leaning and reliability are entangled in the corpus. The regression above controls for the true reliability level, but the two variables are close to collinear in our data, so no analysis

Model	Leaning	Odds ratio	95% CI	p
BERT	Far Left	4.81**	[1.74, 13.34]	0.003
	Slightly Left	2.11*	[1.10, 4.05]	0.025
	Slightly Right	2.11*	[1.05, 4.23]	0.035
	Far Right	6.52***	[2.18, 19.47]	0.001
RoBERTa	Far Left	5.55**	[1.69, 18.26]	0.005
	Slightly Left	1.49	[0.63, 3.51]	0.360
	Slightly Right	2.46*	[1.18, 5.11]	0.016
	Far Right	6.18**	[1.97, 19.37]	0.002
Llama 3	Far Left	6.29**	[1.86, 21.28]	0.003
	Slightly Left	1.41	[0.60, 3.33]	0.436
	Slightly Right	2.74*	[1.23, 6.10]	0.014
	Far Right	6.29**	[1.93, 20.56]	0.002
Mistral	Far Left	4.00*	[1.13, 14.13]	0.031
	Slightly Left	2.12	[0.89, 5.08]	0.090
	Slightly Right	2.56*	[1.18, 5.53]	0.017
	Far Right	5.57**	[1.81, 17.13]	0.003

Table 3: Odds ratios of misclassification by political leaning, relative to the non-labelled group, from a logistic regression controlling for the true reliability level with standard errors clustered on the publisher. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. The intervals are wide: several leaning-by-reliability cells hold fewer than ten publishers (Table 5), which is what limits how far the two variables can be separated.

Model	Accuracy [95% CI]	Tolerant acc. α=1.0 [95% CI]
BERT	0.665 [0.617, 0.708]	0.870 [0.838, 0.900]
RoBERTa	0.690 [0.647, 0.731]	0.875 [0.843, 0.904]
Llama 3	0.665 [0.620, 0.708]	0.879 [0.847, 0.907]
Mistral	0.683 [0.642, 0.727]	0.856 [0.822, 0.888]

Table 4: Publisher-level metrics with cluster-bootstrap confidence intervals (2,000 resamples of the 439 publishers).

can separate them completely. Table 5 counts the publishers in each combination. Of the 207 slightly right publishers, 142 sit in l_1; no slightly left publisher sits there at all, and no far-right publisher reaches l_5. Several cells hold fewer than ten publishers, and two are empty, so the additive control rests on thin support.

Table 6 gives the article-level error rate inside each combination, pooled over the four models. Two cells dominate the corpus and the models get both of them right: the 142 slightly right publishers in l_1 carry a 5% error rate, and the 64 non-labelled publishers in l_3 carry 12%. Every other combination falls between 32% and 99%. The models therefore recognise the two largest blocks of the ground truth, and they do poorly outside them. Cells built on a handful of publishers are noisy, but the composition counts behind them are exact.

Two readings follow. The per-leaning disparity is a property of the corpus and of the model together, not an isolated effect of political leaning. And a system trained on this ground truth inherits the political structure of the dataset, so the disparity would appear in deployment whatever its mechanism.

Why no source-identifier masking. We use the article texts as collected, and we mask no outlet name, byline or boilerplate. Under the publisher-disjoint protocol this cannot inflate the

	Far Left	Sl. Left	Non lab.	Sl. Right	Far Right
l_1	1	—	2	142	7
l_2	6	7	5	22	29
l_3	1	4	64	19	7
l_4	9	29	24	17	9
l_5	3	10	15	7	—
Total	20	50	110	207	52

Table 5: Publishers by political leaning and true reliability level. A dash marks an empty combination.

	Far Left	Sl. Left	Non lab.	Sl. Right	Far Right
l_1	99	—	81	**5**	82
l_2	76	48	91	40	44
l_3	79	93	**12**	79	90
l_4	48	39	32	68	70
l_5	94	73	69	87	—

Table 6: Article-level error rate (%) by political leaning within each true reliability level, pooled over the four models. Bold marks the two combinations that hold most of the corpus. Read these against the publisher counts of Table 5.

results. The model never sees a test-fold publisher during training, so a cue tied to a particular outlet carries no usable information at test time.

7 Lexical Baseline: TF-IDF + Logistic Regression

To separate the contribution of the pre-trained transformers from what surface word statistics already provide, we evaluate a linear lexical baseline under the same publisher-disjoint folds. Each article is represented by TF-IDF features over word unigrams and bigrams, with sublinear term-frequency scaling, a minimum document frequency of five, a maximum document frequency of 0.9 and at most 100,000 features. A multinomial logistic regression with balanced class weights is then trained on these features. The vectoriser is fit on each fold's training articles only, so no test-fold vocabulary leaks into training; article-level predictions are pooled across the five test folds and aggregated to the publisher level by the same majority vote (ties broken toward the lowest level) used for the transformers.

As Table 7 shows, and as the baseline line in the main-paper figures makes visual, the gap between a bag-of-words linear model and the fine-tuned transformers is small, and it closes almost entirely once predictions are aggregated per publisher. Most of the usable signal is therefore lexical, and the choice of encoder matters little for the publisher-level results.

8 Experimental Details

Fold construction. To build the five folds we retain all articles associated with each publisher. Publishers contribute varying numbers of articles, up to 200 each; we include them all to preserve signal diversity rather than discarding potentially informative examples. As a dispersion check we compute, for each reliability class, the standard deviation of its proportion across the five folds. The resulting values range from 1.37 to 3.61 percentage points, indicating that the reliability distribution is well preserved across folds.

	Article level		Publisher level	
Metric	**TF-IDF + LR**	**Transformers**	**TF-IDF + LR**	**Transformers**
Accuracy	0.576	0.583–0.612	0.681	0.665–0.690
Precision (macro)	0.539	0.529–0.560	0.573	0.561–0.617
Recall (macro)	0.523	0.519–0.553	0.581	0.556–0.602
F1 (macro)	0.527	0.523–0.555	0.564	0.551–0.605
Tolerant (0.5)	0.696	0.702–0.723	0.780	0.768–0.782
Tolerant (1.0)	0.816	0.821–0.838	0.879	0.856–0.879

Table 7: The lexical baseline against the range spanned by the four fine-tuned transformers. At the article level the baseline is below all four transformers on accuracy and tolerant accuracy, and level with BERT on the macro metrics. At the publisher level it falls within their range on every metric and ties the best tolerant accuracy (0.879).

Optimisation and batching. Encoders use the `BertTokenizerFast` and `RobertaTokenizerFast` respectively. For the decoders we use the causal `LlamaTokenizer` and `MistralTokenizer` with manual padding management, and enable Flash Attention 2 for Mistral. Batch sizes were adjusted to memory: 32 (training) / 16 (evaluation) for BERT, 16/8 for RoBERTa, 8/8 for Llama with gradient accumulation, and 6/48 for Mistral with five gradient accumulation steps. For Llama 3, early stopping on a stratified 10% hold-out of the training articles, with a patience of two epochs, restores the state with the lowest validation loss; the other models run for a fixed number of epochs. No test fold informs training or model selection, and a fixed global seed (42) ensures identical data splits across architectures.

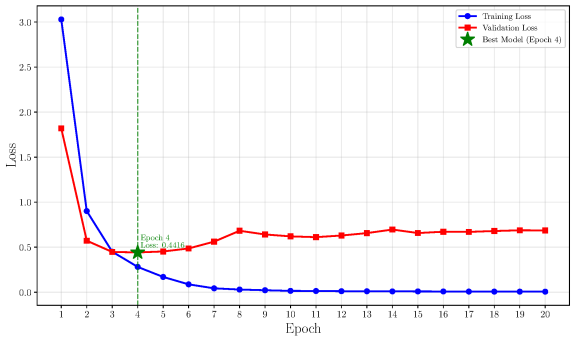
9 Convergence Analysis

To ascertain the most suitable training duration, a preliminary experiment involving 20 epochs was conducted using the Llama 3 model. As can be seen in Figure 7a, the model converges quickly. The validation loss reaches its minimum at epoch 4 (0.4416), after which the generalisation gap widens, indicating overfitting. The empirical evidence gathered from our most complex architecture validates our decision to limit fine-tuning to five epochs across all models, with the exception noted below.

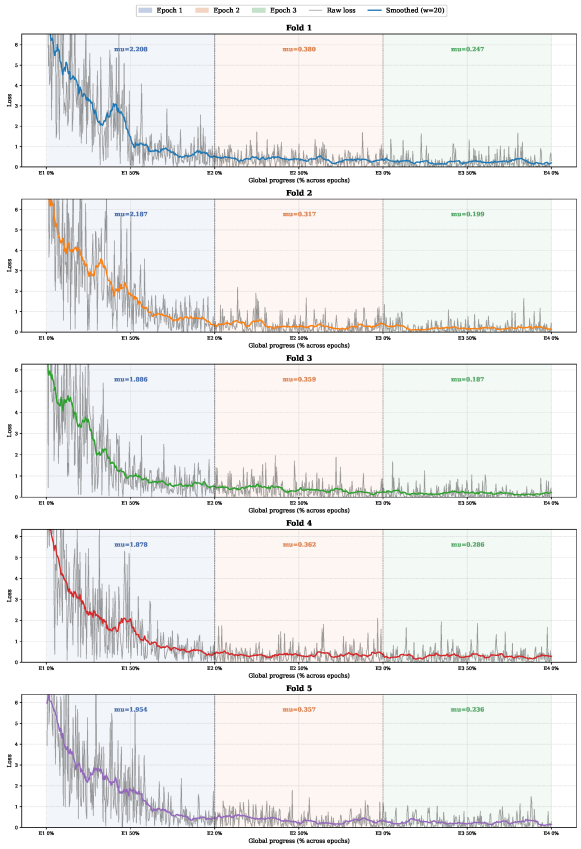
We conducted an analogous convergence check for Mistral across the five-fold cross-validation. As can be seen in Figure 7b, the training loss decreased sharply from an average of 2.023 at Epoch 1 to 0.355 at Epoch 2 and then to 0.231 at Epoch 3. The average improvement between epochs 2 and 3 is 0.124, indicating diminishing returns and stabilisation. This pattern mirrors the rapid convergence observed for Llama 3, supporting the idea of limiting Mistral training to three epochs to improve efficiency without compromising on convergence.

10 Reproducibility

Every figure in this document and in the main paper is regenerated from the stored model predictions by a single notebook, so no figure depends on manual steps or on the order in which cells were executed. The notebook also emits the tables it reports, and validates that all four models are scored against the same ground truth for the same article. Anyone can therefore re-derive the error and bias figures behind the paper's claims about responsible use, instead of relying on the printed versions. The lexical baseline of Section 7 is trained by a separate script in the same repository, and the notebook reads its metrics to draw the baseline line.



(a) Training and Validation loss over 20 epochs for Llama 3. The green marker indicates the optimal model checkpoint at epoch 4. Longer training durations result in poorer generalisation performance.



(b) Training loss over three epochs for Mistral, shown across five folds.